MIL-C-81452B(AS)
16 July 1972
Superseding
MIL-C-81452A(AS)
15 June 1967

MILITARY SPECIFICATION

CABLE ASSEMBLY W-1 of AN/AQS-13

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The Cable Assembly covered by this specification shall connect the AN/AQS-13 hydraulic reeling machine slip rings to the AN/AQS-13 transducer assembly and be capable of withstanding the loads and stresses, without damage or change in performance, incurred during normal operation of the AN/AQS-13 sonar system.

1.2 Associated Equipment - The Cable Assembly shall operate with the sonar systems listed in paragraph 6.6.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

FSC 6150

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-915	Cable, Cord and Wire Electrical (Shipboard use)
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-N-18307	Aeronautical Electronics and Associated Equipment, Nomenclature, and Name Plates For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

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*2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the: Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and material requirements pertaining to connectors shall conform to specification MIL-E-5400.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and material shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of specification MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein.

3.3.1 Reliability -

*3.3.1.1 Operational Stability - The cable assembly shall be capable of operating with satisfactory performance, continuously or intermittently for a period of at least 2500 dip cycles.

*3.3.1.2 Operational Life - The Cable Assembly shall have a total operating life of 2500 dips without servicing or replacement of parts.

3.3.2 Special Purpose Cable -

*3.3.2.1 Cable - Cable materials and utilization shall be in accordance with MIL-C-915, except as modified herein.

*3.3.3 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity. The equipment shall meet the interchangeability requirements of Specification MIL-E-5400 where applicable.

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*3.3.4 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of Specification MIL-STD-461.

*3.3.5 Nomenclatures and Name Plates - Nomenclature assignment and name plate approval for equipment identification shall be in accordance with Specification MIL-N-18307 and as detailed in figure 1 of this specification, or as approved by the procuring activity.

3.3.6 Standard Conditions - The following conditions shall be used to establish normal performance characteristics for making laboratory bench tests.

Temperature	Room Ambient 25 degrees Centigrade ± 5 degrees Centigrade
Altitude	Normal ground level
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity

3.3.7 Service Conditions - The Cable Assembly shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.3.7.1 Vibration - The Cable Assembly shall operate satisfactorily when subjected to the vibration requirements of Curve III of figure II of Specification MIL-E-5400.

*3.3.7.2 Temperature - -40 degrees Centigrade to +55 degrees Centigrade (operating) -60 degrees Centigrade to +85 degrees Centigrade (nonoperating)

3.3.7.3 Altitude - 20 to 500 feet.

3.3.7.4 Shock - 6g.

3.3.7.5 Humidity - 100 percent.

*3.3.7.6 Water Pressure - 500 psia.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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*3.4.1 Operation - The cable shall operate as part of an installed AN/AQS-13, 13A, and 13B sonar systems.

3.5 Detail Requirements

3.5.1 W-1 of AN/AQS-13, Cable Assembly, Special Purpose, Electrical - The Cable shall meet the following requirements.

*3.5.1.1 Function - The Cable Assembly shall connect the hydraulic reeling machine slip rings to the transducer assembly and be capable of withstanding a 1000 pound tension. It shall be capable of being pulled over a 4 inch radius. The sonar cable including connector P7001A with adaptor J7001A (3.5.1.4.7) shall, when mated with the transducer, withstand without leaking and without change in performance a hydrostatic pressure of 500 psi. It shall be capable of withstanding, without damage or change in performance, the "crushing" loads encountered while passing over 4 inch radius sheaves at an average 11 feet per second with a 150 pound load and being loaded on the reel and have a minimum breaking strength of 2000 pounds between the connector P7001A and the cable.

*3.5.1.2 Form Factor - The cable shall be 500 feet long +2 percent -0 percent. The diameter of the cable shall not exceed .565 inch (except the connector assemblies) nor be less than .545 inch. The cable shall not be more than .015 inch out of round and shall be straight within .015 inch within any 4 foot length when measured with a tensional load of 150 pounds. The cable shall not stretch by more than 0.2 percent of its total length and the cable diameter shall remain within the specification limits after stretching has occurred either during tensioning or operation.

*3.5.1.3 Weight - The weight of the cable assembly shall not exceed 115 pounds.

*3.5.1.4 Contents - The cable assembly shall contain the following:

1. Electrical Conductors
2. Shields
3. Center Strength Member
4. Outer Jacket
5. Connectors
6. Strain Relief
7. Adaptor

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*3.5.1.4.1 Electrical Conductors - There shall be in the cable 30 electrical conductors as described in paragraph 3.6 Cable Construction of this specification.

*3.5.1.4.2 Shielding Effectiveness - The fully terminated cable assembly shall have at least 70 db of isolation at 11 kHz between any randomly selected insulated conductor within a bundle to any other randomly selected conductor in any other bundle. (Refer to figure VI.)

*3.5.1.4.3 Strength Member - The strength member shall provide the cable with the capability of supporting a load of 2000 pounds without decreasing the performance specified in 3.5.1.1. It shall be corrosion resistant and be of a construction as specified in 3.6 or equivalent, subject to the approval of the procuring activity.

*3.5.1.4.4 Outer Jacket - The cable outer jacket shall be black polythene or equivalent of a selected grade subject to the approval of the procuring activity. It shall be capable of withstanding a hydrostatic pressure of 500 psi throughout the entire length without leaking. There shall be color coded markings every 50 feet along the entire length of the outer jacket and as specified in figure II of this specification. During extrusion of the outer jacket a yellow longitudinal torque balance reference line shall be extruded integral with the jacket.

*3.5.1.4.5 Connectors - One end of the cable shall be terminated with a connector designated P7001A as described in 3.5.1.4.5.1 of this specification or an equivalent and shall be capable of withstanding a hydrostatic pressure of 500 psi without leaking. It shall be made of a corrosion resistant material as approved by the procuring activity. The strength member shall be mechanically but not electrically connected to the shell of the connector. The other end of the cable shall be terminated with a screw type connector designated P5019B as described in 3.5.1.4.5.1 of this specification or an equivalent and having no positive mechanical connection to the cable except at the solder pots. The strength member and all other shields and wires shall be isolated electrically from the connector shell.

*3.5.1.4.5.1 Cable Connectors - The connection to external circuits shall be provided as follows:

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Reference Designation	Receptacle Type of an Approved Equivalent	Function
P5019B	Screw type with a 28-21P insert. Bendix Scintella Part No. 65-262452-21P	To connect the cable to the reeling machine slip ring assembly.
P7001A	Flange mounted type with a 28-21S insert Cannon Part No. 3106E28-21S-A69-F26-F42. This connector is further described in figure III of this specification. Alternate Bendix Part No. 10-262453-21S.	To connect the cable to the sonar transducer assembly via the adaptor.
J7001A	Adaptor	To connect P7001A to J7001.

*3.5.1.4.5.2 Terminations - The connectors shall be wired as indicated in figure IV of this specification or in a manner as approved by the procuring activity.

*3.5.1.4.6 Strain Relief - A strain relief assembly shall be attached to the cable assembly connector P7001A. It shall prevent excess bending of the cable adjacent to the connector under flight conditions and shall provide a shock absorbing action to cushion the shock of seating the transducer by the cable reeling machine. The top of the strain relief shall be rounded to prevent damage to the aircraft funnel. The strain relief shall be made of corrosive resistant metal or equivalent as approved by the procuring activity.

*3.5.1.4.7 Adaptor - The adaptor shall provide mechanical connection between P7001A and J7001 and meet the mechanical loads and pressure requirements as specified in 3.5.1.1 without failure. The adaptor shall be made of a corrosive resistant material as approved by the procuring activity.

*3.5.1.5 Torque Balance - The cable shall be torque balanced to meet the following requirements. Under the condition of no load the cable with its associated wet end termination shall be suspended from a vertical framework and two pointers attached to the cable at a separation of 15 feet. The pointers shall be lined up in azimuth prior to the application of a 125 pound load. Upon application of the 125 pound load the out of balance torque angle shall not exceed 90 degrees and the restoring torque required to bring the pointers back into azimuth shall not exceed 3 inch pounds.

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*3.5.1.6 Insulation Resistance - All pins on a fully terminated cable assembly except n, p, r, s, j, A, B, C, and Z shall exceed 1000 megohms from each pin to all other pins when measured with a 500 VDC megohmmeter. Pin Z to j shall be 50 megohms minimum. Pin Z to the housing of P7001A shall be 50 megohms minimum. Pins A,B,C to n,p,r,s ; A,B,C to j ; n,p,r,s to j shall be 50 megohms minimum.

*3.5.1.7 Conductor Resistance - All pins on a fully terminated cable assembly shall be a maximum of 13.0 ohms except as follows:

Pin A to A	Maximum	1.2 ohms
Pin B to B	Maximum	1.2 ohms
Pin C to C	Maximum	1.2 ohms
Pin j to j	Maximum	1.0 ohms
Pin n to n	Maximum	2.7 ohms
Pin p to p	Maximum	2.7 ohms
Pin r to r	Maximum	2.7 ohms
Pin s to s	Maximum	2.7 ohms
Pin Z to Z	Maximum	25.0 ohms

*3.5.1.8 Hydrostatic Capability - The cable assembly and wet end termination shall be capable of withstanding application of water pressure of 500 psi for a minimum of 1 hour without leaking.

*3.5.1.9 Reeling Endurance - The cable shall be capable of withstanding, when subjected to temperatures of -40 degrees Centigrade, ambient, and +55 degrees Centigrade, a minimum of 10,000 cycles at each temperature point without failure.

*3.5.1.10 Torsional Oscillation and Whirl - Using a device similar to the one shown in figure V, the cable and wet end terminations shall be capable of withstanding 50,000 cycles of operation without failure, with a load of 250 pounds, a torsional oscillation of + and -180 degrees concurrently with a whirl of 360 degrees at ambient temperature.

*3.5.1.11 Proof Load - The entire cable length and wet end termination shall be capable of withstanding a 250 pound load without failure or change in electrical and physical characteristics. The load shall not be applied to the dry end termination.

*3.5.1.12 Bottom Recovery - The cable and its associated wet end termination shall be capable of withstanding a minimum of six bottom recoveries when the transducer is in a prone position on the bottom of a water filled tank or body of water and then raising the transducer at normal operating speed. A minimum of 10 feet of slack cable shall be used and a minimum of 20 feet of water depth.

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*3.6 Cable Construction - Materials and construction shall be generally in accordance with MIL-C-915 requirements as applicable and with the outline cross section shown in figure VI or an equivalent as approved by the procuring activity.

*3.6.1 Center Strength Member - The center strength member shall have a breaking load in excess of 2000 pounds and shall be insulated with natural low density polythene or an equivalent as approved by the procuring activity.

*3.6.2 Conductors - All conductors, except those specified in 3.6.4, shall consist of unidirectionally stranded plain annealed copper wire. They shall be insulated with .006 inch nominal radial thickness high density polythene or equivalent as approved by the procuring activity. The conductors shall be color coded for identification purposes.

*3.6.3 Conductor Bundles - Using the conductors described in 3.6.2 above, seperate bunules shall be cabled. The lengths of lay of each bundle shall be chosen to minimize cross talk. Power and control leads shall be screened to prevent interference using plain annealed copper wire.

*3.6.4 Transmit Conductors - The conductors for the transmit leads shall consist of unidirectionally stranded plain annealed copper wire. They shall be insulated with .015 inch nominal radial thickness of high density polythene or equivalent. They shall be lapped with plain annealed copper wire and mylar tape.

*3.6.5 Laying Up - The cable shall be laid up in accordance with the cross sectional drawing shown in figure VI or in a similar manner as approved by the procuring activity.

*3.6.6 Outer Jacket - The outer jacket shall consist of medium density grade black polythene or equivalent as approved by the procuring activity. The outer jacket shall be a minimum .05 inch thick at any point on the cable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

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(3) Acceptance Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and the tests as listed in table I of this specification.

*4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

*4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection including equipment already accepted by the government; it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.3 Initial Production Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be used for this test. The cable assembly used in the initial production tests will not be returned to the contractor.

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*4.3.1 Scope of Tests - This equipment may be subjected to those tests listed in table I and any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Monthly Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the tests listed in table I.

*4.4.2 Sampling Tests - Cable Assemblies selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the procuring activity in accordance with the following:

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 25	1
Next 50	1
	1 for each additional 200 or fraction thereof

*Note - If by contract action the initial production test (4.3) is deleted, then a sample test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each cable assembly selected for sampling tests shall be subjected to the test listed in table I.

4.4.3 Monthly Tests - The monthly tests to be performed on one cable assembly per month shall include those listed in table I. These tests shall be conducted only during those months in which a minimum of twenty (20) cables are being manufactured throughout the term of the contract. A minimum of one cable in each production run shall be tested.

4.4.4 Special Tests - Special tests shall be conducted as deemed necessary by the procuring activity for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The cable assembly selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Test Schedule - Selection of cable assemblies for special tests shall be made as follows:

(1) On cable assemblies after an engineering or material change.

(2) Whenever failure reports or other information indicated additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary

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to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary. The tests shall include, but not be limited to, those listed in table I.

4.5.1 Test Data - The test facilities shall keep a daily record of the equipment performance making particular note of any deficiencies or failures. In the event of a failure the log shall indicate:

(1) The cable serial number.

(2) Nature of the failure.

(3) Effect of failure on cable performance.

(4) The number of cycles completed when the failure was first noted.

4.5.2 Failure Report - In the event of a failure the procuring activity shall be notified immediately. A full report shall be submitted to the procuring activity upon completion of the test. In this report, the test facilities shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Presubmission - No item, part or complete cable assembly shall be submitted by the contractor for acceptance or for contractor demonstration tests until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. Complete records shall be required to support compliance with all applicable specifications.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the defects found in the original shall be furnished the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging -

5.1.1 Level A - In addition to the following, Level A packaging shall be in accordance with MIL-P-116 Method IA. The unit package shall contain one cable assembly.

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5.2 Packing - The contents of each exterior container shall consists of one type of equipment or any grouping of equipment as required by the contract or purchase order.

5.3 Marking - In addition to the following, marking shall be in accordance with MIL-E-17555.

5.3.1 Special Markings - None unless otherwise specitied.

5.3.2 Normal Marking - In addtion to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended Use - The equipment covered by this specification is intended for installation as part of the AN/AQS-13, 13A, and 13B sonar systems and shall be utilized to detect, locate, track and communicate with submerged objects in the vicinity of the immersed transducers.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title number and date of this specification.

(2) Selection of applicable levels of packaging and packing (5.1 through 5.3.2).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification where applicable shall be specitically approved in writing by the procuring activity.

(3) Reterenced Specifications - Any reterenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship, to the last previous issue.

6.6 Associated Equipment - The Cable Assembly shall be an integral part of the AN/AQS-13, 13A, and 13B Sonar Systems.

6.7 Serialization - The following methods of serialization shall be used and shall be added to the cable as indicated in figure I of this specification.

Example:

71	09	001	D
Year Cable manufactured	Month Cable manufactured	Serial Number in consecutive order from 1 to 999	Vendor Letters

6.7.1 Vendor Letters - Vendor letters shall be assigned to all qualified vendors and shall be designated on the purchase order or by the procuring activity.

6.7.2 Warning Decal - A warning decal shall be fabricated in accordance with figure VII of this specification and affixed to connector P7001A in accordance with figure VIII of this specification.

6.8 This specification is under the cognizance of AIR-533014B.

Project Number 6105 N152

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Table I

Tests	Preproduction	Initial Production	Sampling	Individual	Monthly
3.5.1.2 Visual and Dimensional	X	X	X	X	X
3.5.1.6 Insulation Resistance	X	X	X	X	X
3.5.1.7 Conductor Resistance	X	X	X	X	X
3.5.1.4.2 Crosstalk	X	X	X		X
3.5.1.8 Hydrostatic Cable	X	X	X	X	X
3.5.1.9 Reeling Endurance	X	X	X		
3.5.1.10 Flight Simulation	X	X	X		
3.5.1.5 Torque Balance	X	X	X	X	X
3.5.1.11 Proof Load	X	X	X		X
3.5.1.12 Bottom Recovery	X	X	X		

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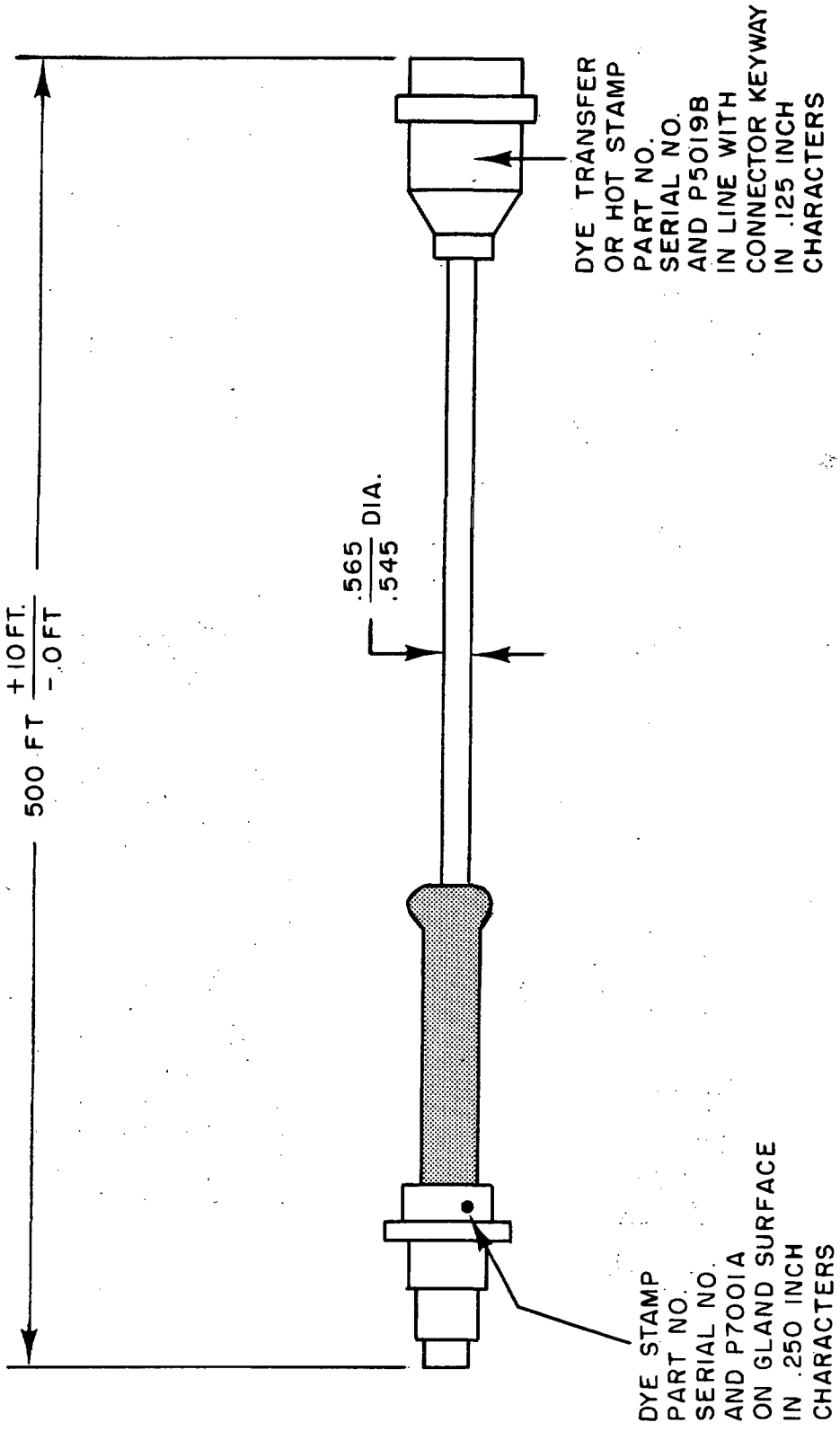


Figure 1
Nomenclature and Nameplates

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1. TOLERANCE ON LOCATION OF COLOR BANDS IS ±1.0 PERCENT OF THE DIMENSION EXCEPT AS SPECIFIED.

2. MARKING MATERIAL SHALL BE VINYL MARKING INK NO. 516, MANUFACTURED BY INDEPENDENT MARKING INK CO., CARDINIA, CALIFORNIA OR EQUIVALENT.

MARKING TABLE	
BAND LOCATION	BAND COLOR (S)
4 FT. 6 IN.	WHITE
7 FT. 6 IN.	YELLOW
21 FT.	RED
50 FT.	GREEN
100 FT.	WHITE
150 FT.	WHITE, GREEN
200 FT.	WHITE, WHITE
250 FT.	WHITE, WHITE, GREEN
300 FT.	WHITE, WHITE, WHITE
350 FT.	WHITE, WHITE, WHITE, GREEN
400 FT.	WHITE, WHITE, WHITE, WHITE
450 FT.	WHITE, WHITE, WHITE, WHITE, GREEN

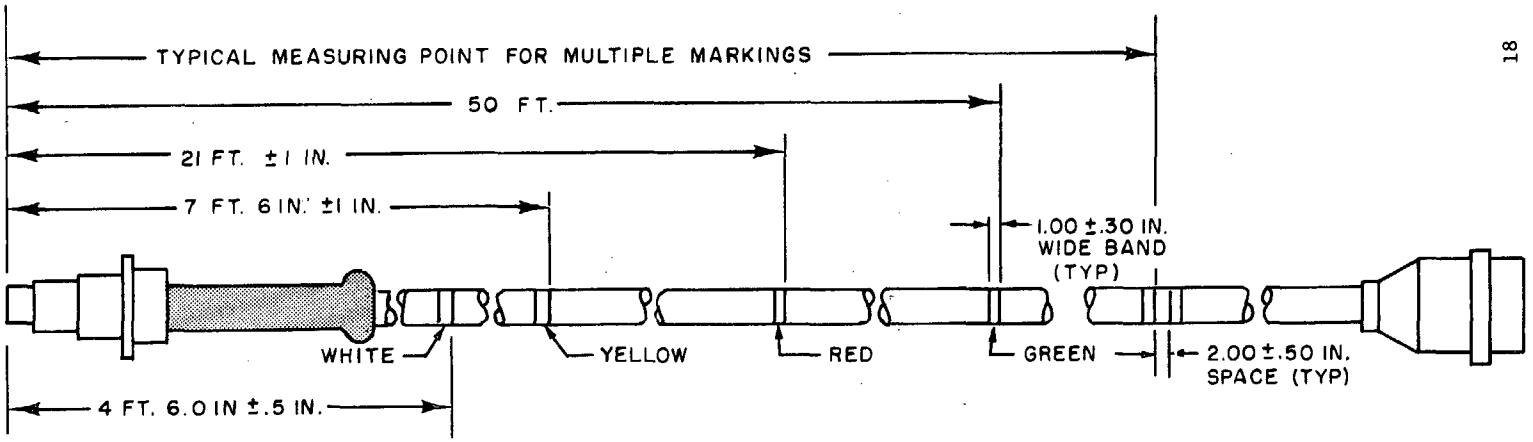


Figure 11
Cable Marking

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Check the source to verify that this is the current version before use.

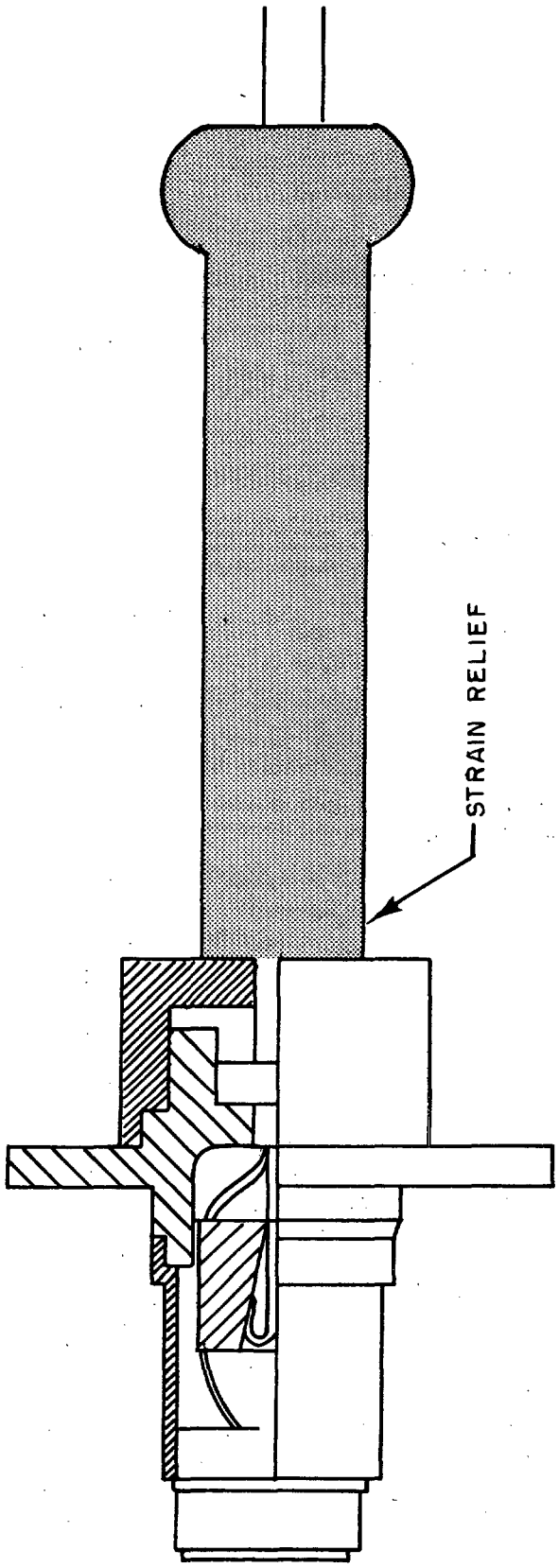
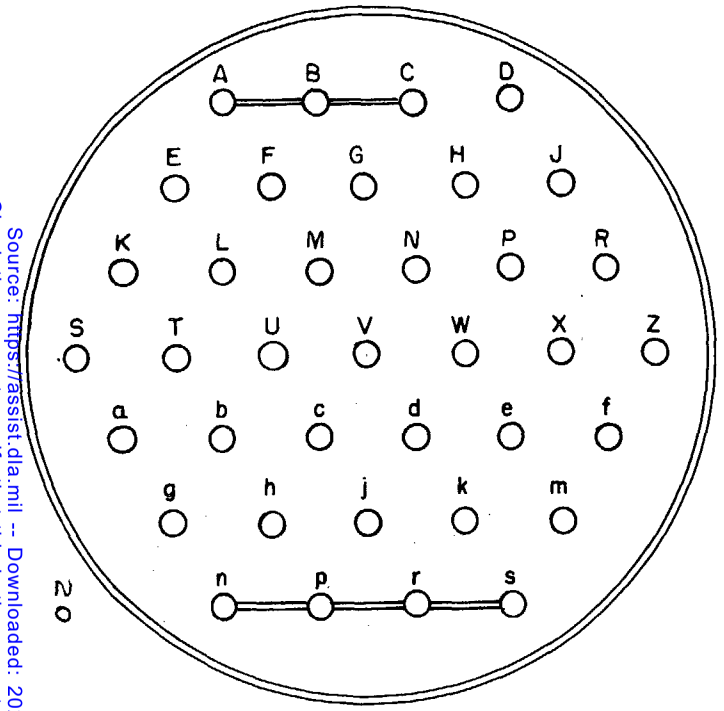
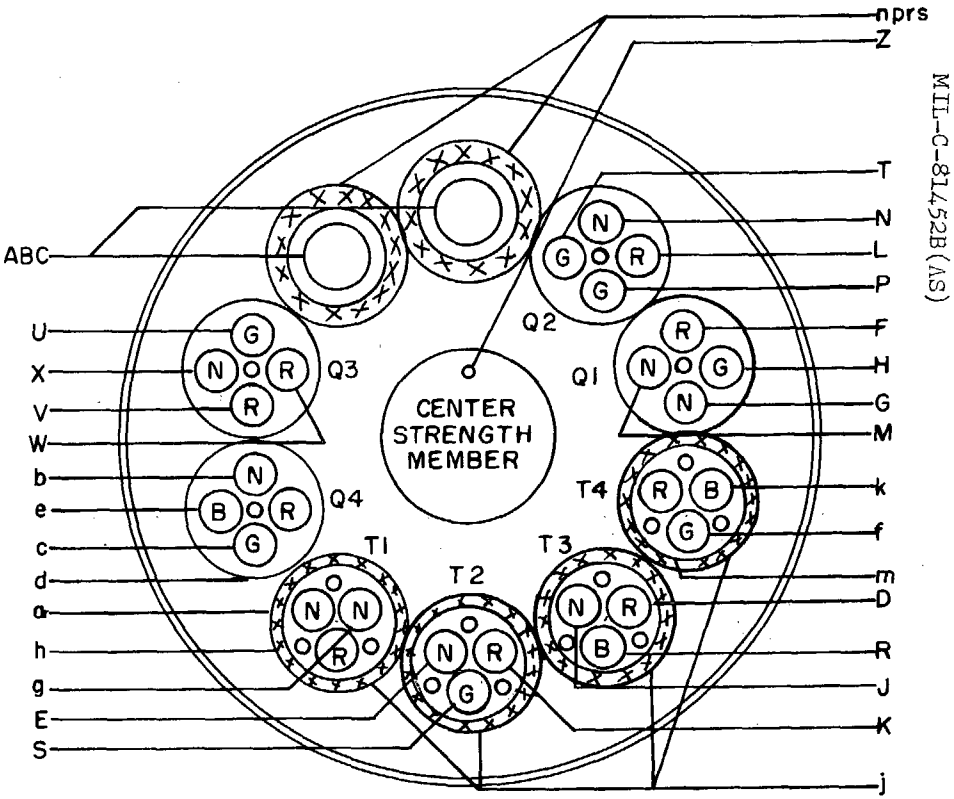


Figure III
P7001A (Typical)

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CONNECTIONS TO PLUG & SOCKET
VIEW IS ON FRONT OF PLUG OR
BACK OF SOCKET



CABLE CROSS-SECTION
OPPOSITE END AS MIRROR IMAGE

(G) - GREEN
(R) - RED
(B) - BLUE
(N) - NATURAL

Figure IV
Lead Termination

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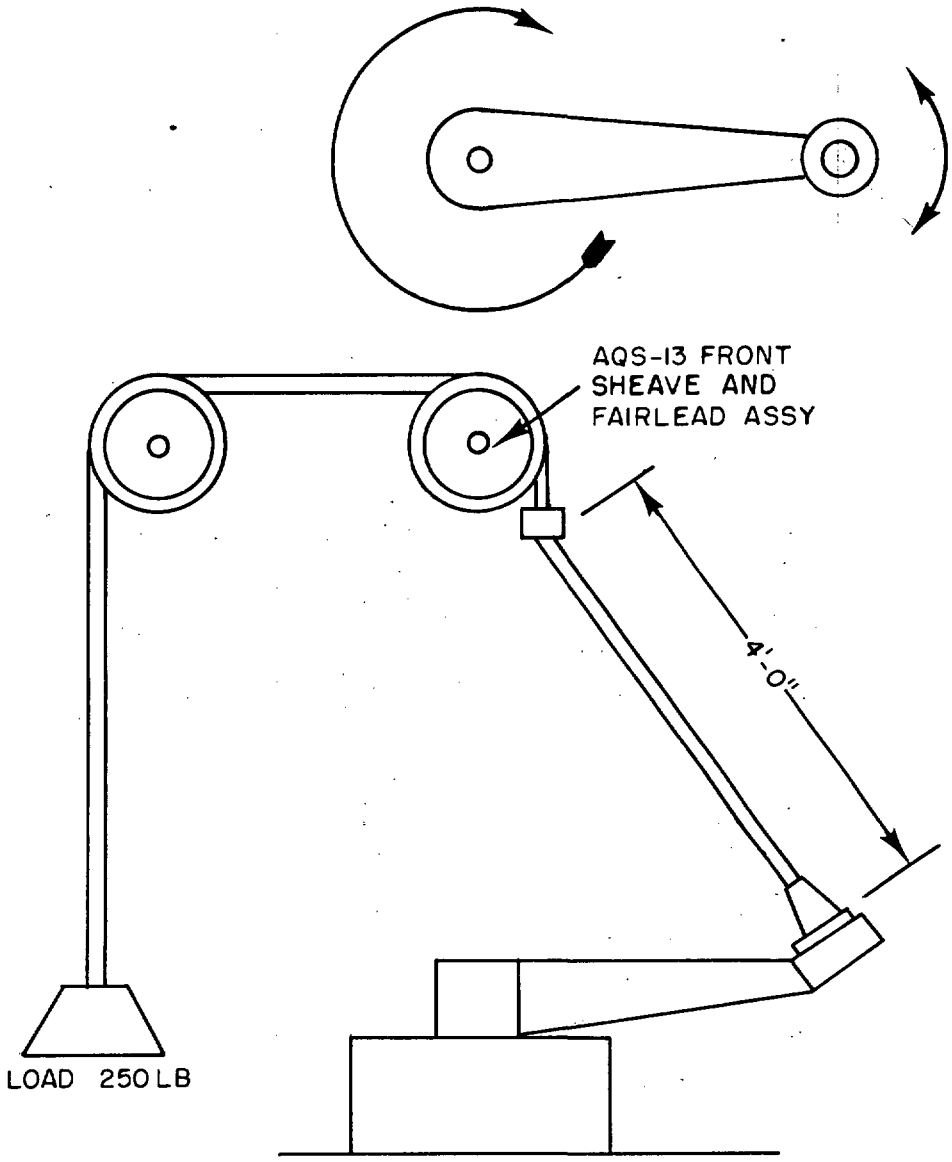


Figure V
Oscillation and Whirl Tester

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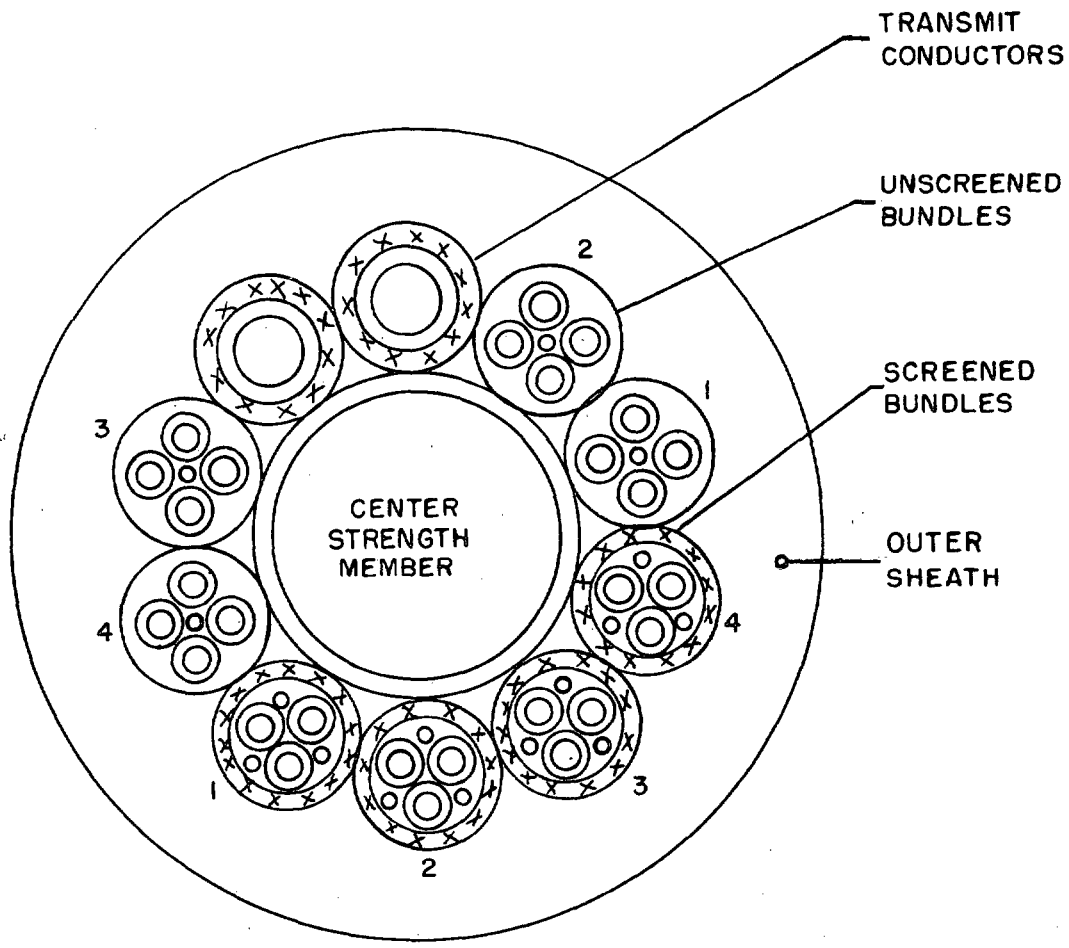


Figure VI
Cross Section Layup (Typical)

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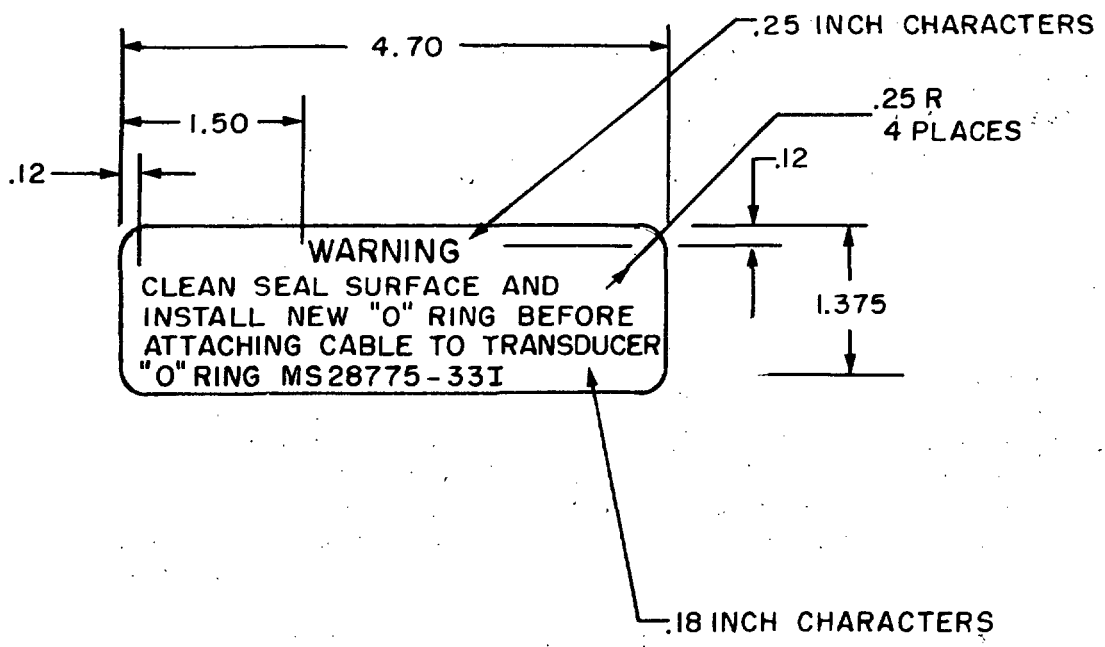


Figure VII
Warning Decal

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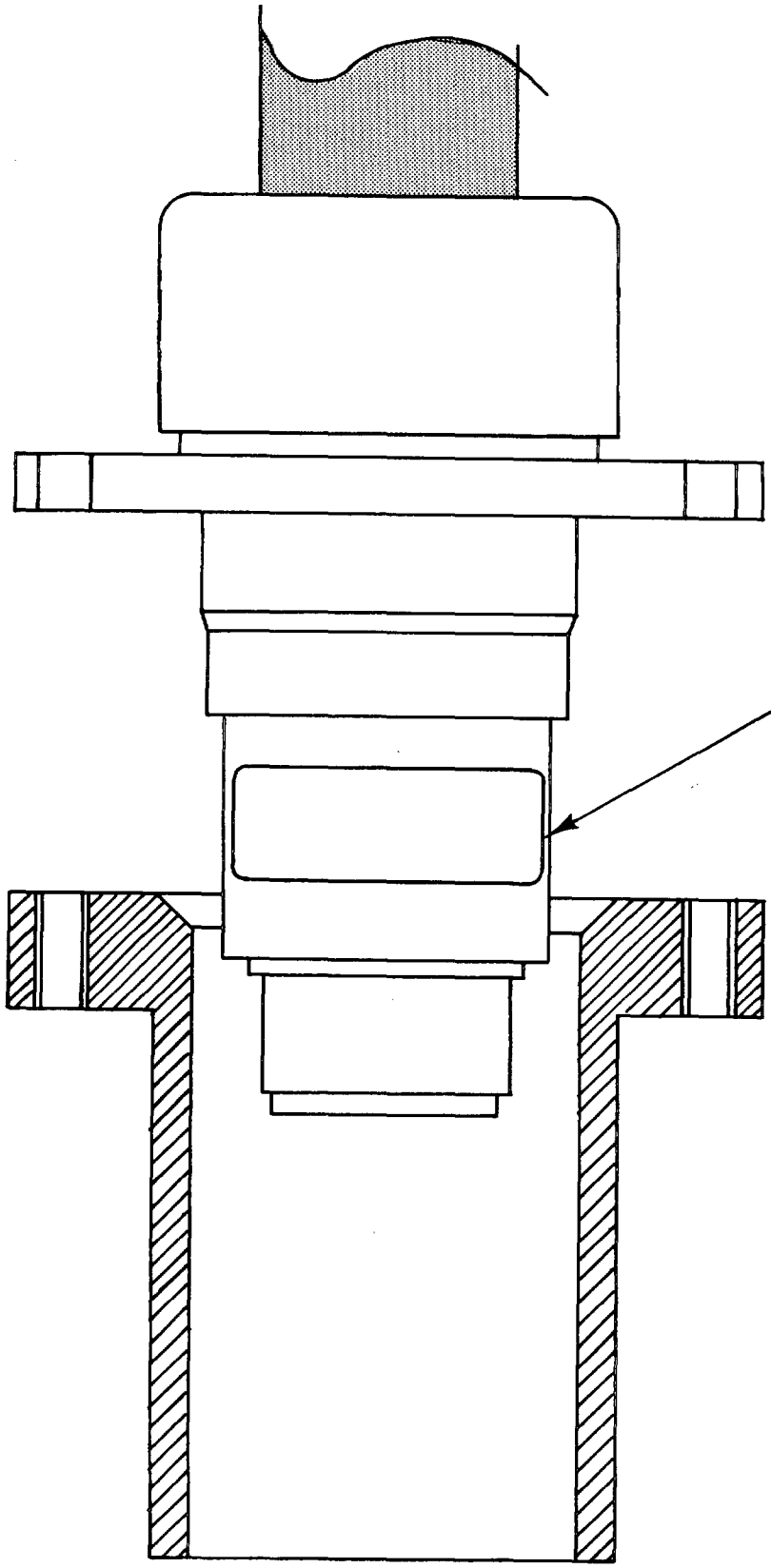


Figure VIII
Decalcomania Placement (Typical)

88

P2

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Naval Air Systems Command
Washington, D.C. 20360

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Naval Air Systems Command (AIR-52021)
Department of the Navy
Washington, D.C.20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81452B(AS) CABLE ASSEMBLY W-1 OF AN/AQS-13

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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